
toli

टोली

An aggregator marketplace for chartered vans,
tempo travellers and buses in India.

Poori toli, ek gaadi.

Product & Engineering Build Plan · Version 1.0

Prepared 10 August 2026 · Customer, Operator and Admin platforms

Mobile-native · Integrated maps · India-first compliance

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0. Name

Toli (टोली) — Hindi for a *band of people moving together*.

Why it works:

- **Semantically exact.** This product does not sell a ride; it sells a group moving as one unit. *Toli* is the closest word in any Indian language to that idea, and no competitor owns it.
- **Warm, not corporate.** It carries the everyday feeling of *bachchon ki toli*, *Holi ki toli* — people who set out together. That is the emotional register of a wedding party, a pilgrimage group, an office offsite.
- **Speaks naturally in a sentence.** "*Toli se book kar liya.*" Two syllables, easy for a driver, a fleet owner, and an app-store listing alike.
- **Clean of collisions.** Not competing for mindshare with RedBus, Chalo, Zingbus, Namma Yatri, Sarathi or Vahan.

Wordmark: **tol**i in lowercase. Tagline: "*Poori toli, ek gaadi.*" (The whole group, one vehicle.) English-market variant: "*Book for the whole group.*"

The one weakness to manage: *tol*i is a Hindi-belt word with weaker instant recognition in Tamil Nadu, Kerala, Karnataka and Andhra/Telangana. Two mitigations:

1. Launch north-and-west first (Jaipur → Delhi NCR → Udaipur → Ahmedabad), which is where your Phase 1 corridor sits anyway. By the time you enter the South, the brand carries its own meaning.
2. Pair the wordmark with a **visual group mark** — a cluster of figures, or a row of seats — so the meaning is legible before the word is understood.

Product naming, applied consistently:

Surface	Name
Customer app	Toli
Operator app + web console	Toli Partner
Driver app	Toli Driver
Internal admin	Toli Admin
Price transparency feature	Toli Fair Price
Corporate product (Phase 3)	Toli for Business

Alternates, held in reserve in case of a trademark conflict:

Name	Meaning / angle
Prayaan	"to set out on a journey" (Sanskrit); travels furthest across India — <i>prayanam</i> in Telugu/Malayalam, <i>prayana</i> in Kannada
Mandali	a troupe or company of people; group-specific in Hindi, Marathi, Gujarati
Rathya	from <i>rath</i> (chariot); premium register, suits the corporate and wedding end
GroupGaadi	plainly descriptive, strong SEO value, zero brand ambiguity

Before you commit, in week 1: a Class 39 (transport and travel arrangement) trademark search on ipindia.gov.in; an MCA company-name reservation; and registration of toli.in, toli.co.in and the closest available .com (a four-letter dictionary-adjacent .com will likely be held by a domain investor — budget for it or pick a compound such as tolitravel.com), plus the handle on Instagram, YouTube and WhatsApp Business. Renaming after launch costs far more than it looks like it will.

1. The single most important strategic decision

Do not build Uber. Build a marketplace with an RFQ engine.

You asked for "Uber-like." That framing will cost you a year if taken literally, because the underlying transaction is fundamentally different:

	Uber / Ola (cabs)	Group charter (vans + buses)
Booking horizon	3 minutes ahead	3 days to 3 months ahead
Unit sold	A seat, right now	A whole vehicle, for a duration
Price formation	Algorithm decides, take it or leave it	Negotiated; operator quotes, customer compares
Trip shape	A → B, 20 minutes	Multi-day, multi-stop, multi-state, driver stays with group
Cost components	One fare	Base + per-km + driver bata + night halt + toll + parking + interstate permit tax
Supply	Individual driver-owners	Fleet businesses with 3–200 vehicles and an office manager
Trust driver	Rating + ETA	Vehicle condition, driver behaviour, whether they show up at 4 AM
Cancellation cost	₹0	₹15,000 wedding disaster

So the product's heart is a **structured quote-and-award engine**, with instant booking layered on top *later* once you have enough rate-card data to price confidently yourself.

The three-lane booking model

Build all three, in this order:

1. **Lane A — Assisted RFQ (launch with this).** Customer posts a requirement → your system fans it out to matched operators → operators respond with structured quotes → customer compares apples-to-apples → books with advance payment. You get liquidity and, crucially, a **pricing dataset**.
2. **Lane B — Instant Book (month 6+).** For high-frequency, well-understood lanes (airport transfers, Delhi–Jaipur, local 8hr/80km packages), you publish a Toli rate and auto-assign to an operator who has pre-committed inventory. This is the Uber-like experience, and it only becomes safe once Lane A has taught you real prices.
3. **Lane C — Contracts (month 10+).** Corporate employee transport, school routes, factory shifts, monthly hotel tie-ups. Recurring revenue, low CAC, and the highest-margin part of this business. Most Indian bus-tech companies eventually discover this is where the money actually is.

Charter vs. seat-selling

Your description ("multi-seat vans and bus for large number of people") is **charter** — the group takes the whole vehicle. That's the right wedge: RedBus owns per-seat intercity ticketing and you will not beat them there. Keep per-seat selling out of scope until Phase 4, where it appears as *empty-leg selling* (an operator returning empty from a wedding sells the seats back) — which is a genuinely differentiated product only an aggregator can build.

2. Who you are building for

Demand side (5 segments, deliberately ordered)

Segment	Trip shape	Why start here / why not
Weddings & family functions	2–5 days, 2–15 vehicles, guest ferrying, high emotion, high budget	Start here. Highest willingness to pay, worst current experience, seasonal but predictable
Corporate offsites & events	1–3 days, invoice-driven, GST invoice mandatory	Start here too. They pay by NEFT, need a proper invoice, low haggling
Pilgrimage & religious groups	4–12 days, price-sensitive, repeat annually, community-organised	High volume, thin margin, but incredible word-of-mouth
School / college trips	Safety-first, compliance-heavy, committee decision-making	Phase 3 — needs verified driver records and parent-facing tracking
Employee daily transport	Recurring monthly contracts	Phase 3–4 — highest LTV, longest sales cycle

Supply side

Indian tempo-traveller and bus operators are **not gig drivers**. They are small businesses. Typical profile: 4–30 vehicles, run by an owner and one dispatcher, currently getting bookings via WhatsApp, phone calls, and a network of travel agents who take 15–25% commission for doing nothing but forwarding a phone number.

Your actual value proposition to them is not "more bookings." It's:

1. **Getting rid of the agent's cut** — you charge less and you're transparent about it.
2. **Guaranteed payment** — agents delay payment 30–90 days. You settle in T+2. This alone will win you supply.
3. **Utilisation** — fill the empty return legs.
4. **A free dispatch tool** — most of them run their fleet on a paper diary. Give them the operator app free and they will stay for the tool even when they're grumbling about commission.

Supply comes first. Do not build a customer app until you have 40 verified vehicles signed in one city. A demand-side app with no supply is a very expensive way to disappoint people.

3. System surfaces

You asked for three logins. You actually need **five surfaces** — the driver is a distinct user from the operator, and operators do their real work on a desktop, not a phone.

#	Surface	User	Platform	Why
1	Toli (customer app)	Group organiser	Android + iOS	The consumer brand
2	Toli Partner (operator app)	Fleet owner / dispatcher	Android + iOS	Quote fast from anywhere; push notifications for new RFQs
3	Toli Partner Console	Operator's office staff	Web (React)	Bulk fleet upload, document management, invoices, settlement reconciliation, roster planning. Nobody manages 40 vehicles on a 6-inch screen
4	Toli Driver	Driver	Android only	Location broadcast, trip start/stop, OTP verification, expense capture, SOS
5	Toli Admin	Your ops, finance, support teams	Web (React)	Verification queue, dispute resolution, manual matching, pricing controls, payouts, fraud review

Why Driver is separate from Partner: the operator does not go on the trip. If you merge them, you'll either give the driver access to commercial data (he'll learn your take rate and disintermediate you) or you'll cripple the operator's app with background location permissions. Separate binaries. This is how Porter, BlackBuck and every serious Indian fleet platform is built.

Why Driver is Android-only: ~99% of commercial drivers in this segment are on Android, usually a sub-₹12,000 device with 3GB RAM and an aggressive OEM battery killer (Xiaomi, Vivo, Oppo, Realme). Building iOS here is wasted effort; building it *natively in Kotlin* is not optional, for reasons in §5.

4. Feature scope by surface

4.1 Customer app (Toli)

Onboarding

- Phone + OTP only (no password). Use a DLT-registered SMS provider with WhatsApp OTP fallback — Indian SMS delivery is genuinely unreliable
- Optional profile: name, email, GSTIN (critical — corporates cannot expense a trip without a GST invoice)
- Language switcher: English, Hindi, Marathi, Tamil, Telugu, Kannada, Bengali, Gujarati, Punjabi (start with English + Hindi, ship the i18n plumbing on day one)

Requirement builder — the most important screen in the product

- Trip type: One-way / Round trip / Multi-day tour / Local package (8hr–80km, 12hr–120km) / Airport transfer / Recurring
- Pickup + drop with map pin, plus intermediate stops (drag to reorder)
- Date, time, return date
- Passenger count → suggests vehicle configuration (see §4.6)
- Vehicle preferences: AC/non-AC, seat count, luggage carrier, pushback seats, sleeper, wheelchair-accessible (now a regulatory expectation under MVAG 2025)
- Extras: driver in uniform, decorated vehicle (wedding), music system, hostess, first-aid, live tracking link for guests
- Free-text note + photo/voice-note attachment (huge for Indian users — they'd rather record 20 seconds than fill a form)
- **Save as template** for repeat trips

Quote comparison

- Cards showing operator rating, vehicle photos (real ones, uploaded and verified — not stock images), year of manufacture, **all-inclusive price with the extras broken out**
- Standardised inclusion/exclusion chips: **Toll included** **Parking excluded** **Driver bata ₹500/day** **Interstate tax included** **Night halt ₹300**
- **This is your core UX innovation.** The reason group booking is miserable today is that quotes are not comparable. Force every quote into the same schema and you've already won.
- Chat with operator (in-app, phone number masked via a cloud-telephony number)
- One-tap counter-offer

Booking & payment

- Advance (20–30%) online, balance either online before departure or cash to driver
- Payment methods in priority order: **UPI intent/collect** (70%+ of transactions), cards, netbanking, wallets, and *Pay Later* / *NEFT with PO* for corporates
- Cancellation policy shown *before* payment, tiered by days-to-departure
- Auto-generated GST invoice as PDF, emailed and downloadable

Live trip

- Vehicle + driver details released T-12h (photo, name, phone, RC number, DL verified badge)
- Live map tracking

- **Shareable public tracking link** — no app install, works in any browser. For a wedding, 60 guests want to know where the bus is. This is your best organic acquisition channel; put a soft "Book on Toli" CTA on that page
- SOS button → your 24×7 ops desk + configured emergency contacts
- In-trip issues: "AC not working", "driver rash", "vehicle different from booked" → routed to ops with SLA

Post-trip

- Structured rating: vehicle cleanliness, driver behaviour, punctuality, matched-the-booking
- Photo evidence for disputes
- Invoice, GST download, rebook-in-one-tap

4.2 Operator app + console (Toli Partner)

Onboarding & KYC

- Business PAN, GSTIN, address proof, cancelled cheque / bank details, signed agreement (e-sign via Aadhaar eSign or DocuSign)
- Per-vehicle: RC, permit (state tourist / **AITP** for interstate), fitness certificate, insurance, PUC, VLTD/AIS-140 certificate, 6+ real photos (exterior, interior, seats, boot)
- Per-driver: DL, Aadhaar/police verification, photo, medical fitness declaration
- **Auto-verify against government sources:** VAHAN for RC, permit and fitness; Sarathi/Parivahan for DL; GSTIN API for tax registration. Manual review only for exceptions. This is your defensibility — most competitors accept a photo of a document and hope.
- Expiry tracking with 30/15/7-day reminders, and **hard auto-suspension** of a vehicle whose permit or insurance has lapsed. Non-negotiable.

Lead flow

- Push notification within seconds of a matching RFQ
- Structured quote form: base fare, per-km rate, min-km/day, driver bata, night halt, toll/parking inclusion toggle, interstate tax inclusion toggle, extras
- **Rate card mode** — operator sets standing rates by vehicle type and route band; system auto-generates quotes on their behalf and they simply confirm. Dramatically raises response rate, which is the metric that kills marketplaces
- Response-time leaderboard and visible impact on ranking

Fleet operations

- Calendar/gantt view of vehicle availability; block dates for maintenance
- Assign vehicle + driver to a confirmed booking
- Trip lifecycle: assigned → driver dispatched → started (odometer photo) → in transit → completed (odometer photo)
- Sub-contracting: mark a trip as passed to a partner operator, with liability trail. **This happens constantly in the real market** — pretending it doesn't will cause your worst incidents. Model it explicitly and require the sub-contracted vehicle to also be on-platform

Money

- Earnings dashboard: pending, in-escrow, settled
- Settlement statements with TDS and commission line items
- Instant-settlement option at a small fee (a real revenue line, and operators want it badly)
- Invoice download for their own accounting

4.3 Driver app (Toli Driver)

Deliberately minimal — three big buttons, works on a bad phone, works offline.

- OTP login, today's trip card
- Navigate (hands off to Google Maps/Mappls app rather than reinventing turn-by-turn)
- Start trip: OTP from passenger + odometer photo + vehicle-condition photo
- Background location broadcast (foreground service, ~10s ping while active, 60s while idle)
- Expense capture: toll, parking, fuel, state permit — photo of receipt, auto-added to trip settlement
- SOS
- Offline queue: all events buffer locally and sync when connectivity returns. Half your trips will pass through zones with no data
- Language: full Hindi/regional, icon-heavy, minimal text

4.4 Admin console

- **Verification queue** — document review with side-by-side government API result
- **Booking control tower** — every live trip on one map, with delay/deviation/SOS alerts
- **Manual matching desk** — for the first 12 months, a human will beat your algorithm on high-value RFQs. Build the tooling to let them
- **Dispute & refund** workflow with evidence timeline
- **Pricing controls** — commission by segment/city/operator tier, promo engine, surge caps
- **Finance** — payout runs, reconciliation, GST reports, TDS/TCS registers
- **Fraud** — off-platform leakage detection (see §10), duplicate accounts, GPS spoofing
- **CMS** — cities, vehicle types, cancellation policies, banners, push campaigns
- **RBAC + full audit log** — every admin action attributable. Regulators and your future auditors will ask

4.5 Notifications

WhatsApp Business API is your primary channel, not push. Indian users disable push and don't open apps between trips, but they read WhatsApp. Budget for it: booking confirmations, driver details, tracking links, payment reminders, invoices. SMS via DLT-registered templates as fallback. Push for in-session urgency only.

4.6 Vehicle taxonomy (get this right in the schema on day one)

Class	Seats	Typical use
MPV / SUV	6–7	Small family group
Force Traveller / Tempo Traveller	9, 12, 13, 17, 20, 26	The workhorse of this market
Mini bus	21, 25, 32	Local functions, school
Coach — seater	35, 40, 45, 49	Corporate, pilgrimage
Coach — Volvo/Scania multi-axle	41–49	Premium long distance
Sleeper coach	30–40 berths	Overnight
Double-decker / open-top	varies	Events, tourism

Attributes: AC / non-AC, push-back, luggage carrier, LED/TV, mic system, washroom (rare), year of manufacture, BS emission norm, fuel type, wheelchair accessibility.

5. Technology stack

5.1 Mobile — you said native, here's the honest recommendation

Surface	Recommendation	Reasoning
Customer app	Flutter	Rich UI, one codebase for Android + iOS, near-native performance, excellent Google Maps plugins, large and affordable talent pool in India
Operator app	Flutter (shared design system with customer)	Same team can build both
Driver app	Native Kotlin (Android only)	Non-negotiable
Web consoles	React + TypeScript + Vite, TanStack Query, shadcn/ui	Standard, fast to hire

Why native Kotlin specifically for the driver app. This is where "mobile native" genuinely matters. Reliable background location on Indian budget Android is a fight against OEM battery optimisation, Doze mode, and process death. You need `ForegroundService` with `FOREGROUND_SERVICE_LOCATION`, a `FusedLocationProviderClient` tuned to balance accuracy against battery, a `WorkManager` sync queue, a Room-backed offline buffer, and OEM-specific autostart-permission prompts (Xiaomi's MIUI in particular). Every cross-platform abstraction leaks here, and when it leaks your customer doesn't see the bus on the map — which is the one thing they're paying you for. Also: your APK must stay under ~15MB for install conversion on cheap devices.

If you insist on fully native everywhere (Kotlin + Swift for all three apps): expect roughly +60–70% mobile engineering cost and +3 months to MVP for a marginal UX gain on the customer side. My recommendation is to spend that budget on supply acquisition instead. Revisit fully-native for the customer app at Series A scale.

5.2 Backend

```
API Gateway (Kong / AWS ALB)
|
+---+-----+
| Core Platform – NestJS (TypeScript)      |
| Modular monolith, module boundaries enforced |
| identity · catalog · rfq · quoting · booking|
| pricing · payments · settlement · ratings  |
| notifications · compliance · admin        |
+---+-----+
|
+---+-----+ +-----+ +-----+
| Tracking Svc | | Matching Svc | | Geo Svc |
| Go – high write | | Python – scoring | | Go – OSRM, |
| GPS ingest, WS | | ranking, ETA ML | | geocode cache|
+-----+ +-----+ +-----+
```

Data: PostgreSQL 16 + PostGIS (primary) · Redis (cache, locks, geo)
TimescaleDB or ClickHouse (GPS history, analytics)
S3-compatible object store (documents, photos)
Redpanda/Kafka (event bus – from Phase 2)

Start as a modular monolith. Microservices at seed stage are a tax you cannot afford. Enforce module boundaries in code (NestJS modules, no cross-module DB access, events between modules) so you can split later when a specific module actually needs independent scaling. The only services worth extracting on day one are **tracking** (very different write profile — thousands of GPS points per second at scale) and **geo/routing** (CPU-bound, wants its own machines).

Why NestJS/TypeScript: shared DTOs with the React consoles, opinionated structure that survives team turnover, huge and inexpensive Indian talent pool. Go is a defensible alternative for the whole backend if your CTO prefers it — pick one and don't split the team's mental model.

5.3 Infrastructure

- **Cloud:** AWS Mumbai (ap-south-1) or GCP Mumbai. Data localisation matters for RBI payment data and DPDP compliance
- **Compute:** ECS Fargate or GKE Autopilot. Do not run your own Kubernetes control plane at this stage
- **CI/CD:** GitHub Actions → staging → prod. Fastlane for app store deploys
- **Observability:** OpenTelemetry → Grafana Cloud or Datadog; Sentry for mobile crashes; structured JSON logs
- **Feature flags:** Unleash or GrowthBook — you'll want city-by-city rollout
- **Environments:** dev, staging, prod. Seed staging with realistic Indian data, not **John Doe**

5.4 Estimated infra cost

Stage	Monthly
MVP, 1 city, <500 trips/mo	₹45,000 – ₹80,000
5 cities, ~5,000 trips/mo	₹2.5L – ₹4L
20 cities, ~40,000 trips/mo	₹12L – ₹20L

Maps and SMS/WhatsApp will be a larger share of this than compute. Budget accordingly.

6. Maps and location — the India-specific decisions

This deserves its own section because naive choices here will either bankrupt you or give you wrong ETAs.

6.1 Provider strategy: use three, abstract all of them

Build a `MapProvider` interface from day one with implementations you can swap per-capability and per-region. Never let provider SDK types leak into your domain code.

Capability	Recommendation
Map rendering (mobile)	Google Maps SDK. Google cut Maps Platform prices for India-billed customers by up to 70% in 2024, bills in rupees, and the mobile Maps SDK sits in the free tier. Familiar to every Indian user
Places autocomplete / geocoding (urban)	Google Places , with session tokens and aggressive debouncing (fire at ≥ 3 chars, 300ms)
Geocoding (rural, village-level, non-standard addresses)	Mappls (MapmyIndia). Materially better on village names, landmark-based addresses, and small-town road networks — exactly the destinations a pilgrimage or wedding charter goes to. Their data underpins Indian government and ISRO systems. Use as fallback when Google confidence is low
Bulk routing / distance matrix for quoting	Self-hosted OSRM or Valhalla on an OpenStreetMap India extract. This is the big one — see below
User-facing ETA and live traffic	Google Directions , called sparingly
Turn-by-turn navigation	Deep-link out to Google Maps or Mappls Navigation. Do not build in-app navigation

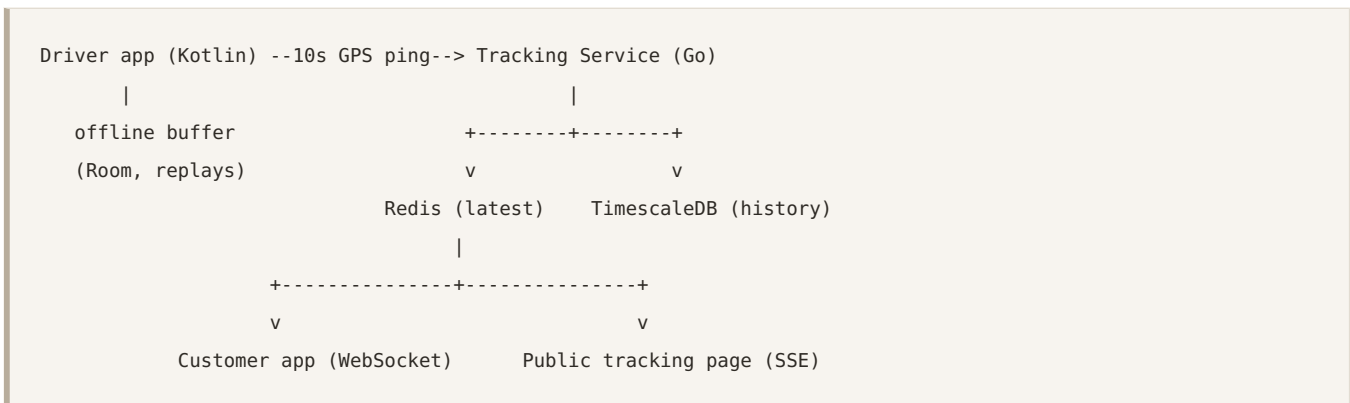
6.2 Why self-hosted routing is essential

Every RFQ fans out to 10–30 operators. Every quote needs a distance and duration estimate. Every price validation needs one. Every "is this operator near enough" check needs one. You will burn **hundreds of thousands of routing calls per month** on internal computation that the user never sees.

Run OSRM on the India OSM extract (~1.5GB PBF) on a single 16GB instance. Cost: about ₹8,000/month. The same volume on a commercial API: ₹4–8 lakh/month. Use commercial APIs only for the numbers a user actually looks at.

Cache aggressively: geocode results forever, route distances for common origin-destination pairs with a 30-day TTL, keyed on rounded coordinates.

6.3 Live tracking architecture



- Snap GPS to road before display (raw pings on Indian roads look drunk)
- Detect: deviation from planned route, unexplained stops >20 min, overspeeding, harsh braking (accelerometer), night driving hours
- **Ingest AIS-140 VLTD feeds too.** All commercial passenger vehicles in India must carry an AIS-140-compliant tracking device with panic buttons. Many operators already have a telematics vendor. Ingesting that feed gives you tracking even when the driver's phone dies — which it will. Build a webhook/pull adapter layer for the common Indian VLTD vendors
- Retain GPS history for at least 90 days for dispute resolution and safety investigation

6.4 ETA quality

Google's ETA is tuned for cars. A 26-seat tempo traveller on a ghat road, or a 45-seat coach on a state highway, is meaningfully slower and stops more. Collect your own actual-vs-predicted data from trip one and train a correction model per (vehicle class × road class × time of day) once you have ~5,000 trips. **On-time arrival is the single trust metric this business lives on** — a wedding party stranded because your ETA was optimistic is a brand-ending event.

7. The pricing engine

The heart of the marketplace. Model it explicitly rather than letting operators send free-text prices.

7.1 Canonical quote schema

```
Quote {
  base_fare           // for local packages: covers min hrs + min km
  included_km         // e.g. 80 km
  included_hours      // e.g. 8 hours
  extra_km_rate       // ₹/km beyond included
  extra_hour_rate     // ₹/hr beyond included
  per_km_rate         // for outstation: ₹/km, on a MINIMUM DAILY KM basis
  min_km_per_day      // typically 250-300 km/day – critical, often hidden
  driver_bata_per_day // driver allowance, ₹300-800
  night_halt_charge   // ₹300-1000 per night
  toll_included       : bool
  parking_included    : bool
  state_permit_included : bool // interstate taxes; can be ₹5000+ per state
  fuel_included       : bool // almost always true; flag rare exceptions
  gst_rate            // see §8.3
  estimated_total      // computed, shown prominently
  worst_case_total     // computed, shown honestly
  cancellation_policy_id
  validity_until       // quotes expire – 24-48h
}
```

Making **min_km_per_day** and **state_permit_included** mandatory, visible fields is your product's single biggest customer-trust win. These are the two charges that produce the classic Indian charter experience of a ₹28,000 quote turning into a ₹41,000 bill at the end of the trip.

7.2 Toli Fair Price band

Once you have ~2,000 quotes, compute a p25–p75 band per (route band × vehicle class × season) and show it on the comparison screen: “Quotes for this trip usually fall between ₹24,000 and ₹31,000.” This is high-trust, easy to explain, and it disciplines operators without you setting prices (which would attract MVAG fare-regulation scrutiny you don't want yet).

7.3 Seasonality

Indian group travel is violently seasonal: wedding *muhurat* dates (Nov–Feb, Apr–Jun), school holidays, Char Dham yatra season (Apr–Oct), Diwali, Ganesh Chaturthi, exam-result travel. Build a **calendar-driven demand multiplier** into the pricing model and, more importantly, into your **supply planning** — on a peak wedding date, every tempo traveller in Jaipur is already booked, and your job is to have secured inventory two months earlier.

7.4 Take rate

Start at **8–12% commission** on trip value, charged to the operator, and be loudly transparent about it. Reference points: travel agents currently take 15–25%; MVAG 2025 sets the driver's minimum share at 80% for driver-owned vehicles,

which effectively caps aggregator take at 20% in regulated ride-hailing. Undercutting agents visibly is your supply pitch. Additional revenue lines: instant settlement fee, featured placement, insurance attach, corporate SaaS subscription for the console.

8. Money flow, tax, and compliance

Engage a chartered accountant and a transport-law counsel in month 1. The items below are the structural decisions that shape your database schema and settlement code, so they cannot be deferred — but treat this section as the agenda for that conversation, not as legal advice.

8.1 Payment stack

- **Gateway:** Razorpay or Cashfree. Both offer marketplace split-settlement products (Razorpay Route, Cashfree Easy Split) that let you hold funds and release to operator sub-accounts — essential for advance/balance flows
- **Payouts:** RazorpayX or Cashfree Payouts, IMPS/NEFT/UPI to operator bank accounts
- **UPI is the default.** Support intent flow (deep-link to GPay/PhonePe/Paytm) and QR. Cards are a minority
- **Corporate:** invoice + NEFT with credit terms, PO reference field, monthly consolidated billing
- **Cash-to-driver balance:** model it. Driver marks cash collected in-app, it's reconciled against the operator's settlement. Half your first-year balance payments will be cash

8.2 The escrow question

You are not an RBI-licensed payment aggregator, so **do not hold customer money in your own current account** as a matter of routine — use the gateway's escrow/split product where funds sit in a nodal account until release. Release trigger: trip completion + 24h dispute window, or immediately for verified Tier-1 operators.

8.3 GST — the structural issue you must resolve before writing settlement code

Under **Section 9(5) of the CGST Act**, an electronic commerce operator can be *deemed to be the supplier* for notified services, including passenger transport supplied through the platform — meaning the platform, not the operator, pays GST on the whole fare, not merely on its commission. Notification 17/2017-CT(Rate) as amended by 16/2023 governs which vehicle categories fall in. Rates in play for passenger transport are typically 5% without input tax credit or 12% with ITC, while *rental of a vehicle with operator* is treated differently again at 18%.

Whether your charter product is "transportation of passengers" or "rental of a motor vehicle" changes your tax rate, your ITC position, and your unit economics by 7 percentage points. Get a written opinion. Then build the invoicing engine to handle both, with the applicable treatment as a configurable attribute per booking type.

Also in scope:

- **TCS under Section 52 CGST** (1% by ECOs) and **TDS under Section 194-O of the Income Tax Act** (1% on e-commerce participants) — both need automated deduction, deposit, and certificate generation
- **E-invoicing** thresholds, if applicable
- Operators below the registration threshold: Section 23(2) allows exemption from compulsory registration for small suppliers selling through an ECO paying under 9(5) — this matters a lot, because many of your smaller operators are unregistered

8.4 Motor Vehicle Aggregator Guidelines, 2025

MoRTH issued revised guidelines on 1 July 2025, superseding the 2020 framework, with states advised to adopt them from 1 October 2025. Key obligations if you fall in scope:

- **Licence from the State Transport Authority** under Section 93 of the Motor Vehicles Act, per state you operate in. A central single-window portal is planned. Applicant must be a company, LLP, cooperative society, or a compliant partnership firm — so **incorporate a Private Limited company early**
- **Security deposit** (bank guarantee or insurance surety bond)
- **Fare regulation:** dynamic pricing permitted within a band of 50% below to 2× the state-notified base fare; drivers who own their vehicle get at least 80% of the fare (60% for aggregator-owned)
- **Cancellation fee capped at ₹100** where cancellation lacks a valid reason, for either side
- **Driver requirements:** police verification, medical test, psychological assessment before onboarding; induction training plus annual refreshers
- **Grievance officer** appointed, with contact details displayed in-app
- **Divyangjan-accessible vehicles** required in the fleet mix; states may set EV adoption targets
- **Penalties** from ₹1 lakh to ₹1 crore; repeat violations mean 3-month suspension and eventual cancellation

Important nuance to get counsel on: the guidelines expressly exclude certain travel service portals and interoperable networks that do not onboard drivers and vehicles. A pure charter marketplace where the *operator* contracts with the passenger may sit differently from a ride-hailing aggregator. Your entity structure and terms of service should be designed with this in mind from the start, not retrofitted.

8.5 Vehicle and permit compliance

- **All India Tourist Permit** under the All India Tourist Vehicles (Permit) Rules, 2023 for interstate charter. For a tempo traveller, expect roughly ₹15,000–₹25,000 in permit fees plus per-state composite tax. Your platform must **block interstate bookings for vehicles without a valid AITP** — an operator caught without one faces up to ₹10,000 fine and vehicle detention under s.192A, and your passengers are stranded at a border check post at 2 AM
- **Vehicle age limits** apply (AITP typically not granted past 12 years from first registration; stricter in Delhi NCT for diesel). Encode as validation rules
- **AIS-140** VLTD device with panic buttons — mandatory for public service vehicles
- **Fitness certificate, commercial insurance, PUC** — track expiry, auto-suspend

8.6 Data protection

The **Digital Personal Data Protection Act, 2023** applies. Location data of passengers, driver Aadhaar/DL data, and payment identifiers are all sensitive. Requirements: explicit purpose-limited consent, a consent-withdrawal mechanism, breach notification, data retention limits, a Data Protection Officer once you cross scale thresholds. Practical build items: consent ledger table, PII encryption at rest, phone-number masking in all logs, per-field access control in the admin console, right-to-erasure workflow.

8.7 Insurance

- Verify each vehicle's commercial passenger insurance at onboarding and at renewal
- Offer per-passenger trip insurance as an attach product (revenue line, and a genuine trust signal for parents booking school trips)
- Carry your own professional indemnity and platform liability cover

9. Data model — core entities

```
User +- CustomerProfile
      +- OperatorProfile +- Vehicle +- VehicleDocument (type, number, expiry, verification_status)
      |                   |          +- VehicleAvailability (date ranges, blocks)
      |                   +- Driver  ---- DriverDocument
      |                   +- RateCard (vehicle_class, route_band, season, rates...)
      |                   +- BankAccount
      +- AdminUser ---- Role ---- Permission

TripRequest (RFQ)
  +- Itinerary ---- Stop[] (sequence, place_id, lat/lng, arrival_window, halt_duration)
  +- VehicleRequirement[] (class, count, ac, features)
  +- ExtraRequirement[]
  +- Quote[] ---- QuoteLineItem[] // the §7.1 schema
    +- Booking
      +- Payment[] (advance, balance, refund; mode, gateway_ref, status)
      +- Assignment[] (vehicle, driver, sub_contracted_to?)
      +- TripExecution --+- TripEvent[] (started, stop_reached, completed, deviation, sos)
      |                   +- LocationPing[] → TimescaleDB
      |                   +- TripExpense[] (toll, parking, permit; receipt_url)
      +- Invoice ---- InvoiceLine[] (with GST breakup, SAC code)
      +- Settlement (gross, commission, tds, tcs, net, status, utr)
      +- Review[]
      +- Dispute[] ---- DisputeEvidence[]

ComplianceCheck (entity_type, entity_id, source: VAHAN|SARATHI|GSTN, result, checked_at)
AuditLog (actor, action, entity, before, after, ip, timestamp)
ConsentRecord (user, purpose, granted_at, withdrawn_at)
```

Design notes:

- Money in **paise as integers**, never floats
- All timestamps UTC, render in IST
- Soft-delete everything financial; hard-delete only for DPDP erasure requests, with a tombstone
- **Vehicle** has a lifecycle state machine: **draft** → **pending_verification** → **active** → **suspended** → **retired**, with suspension reasons enumerated
- Every price shown to a user is snapshotted at display time — never recompute a historical price

10. The problems that will actually hurt you

Disintermediation (leakage)

Your biggest existential threat. Customer and operator connect once, then transact directly on WhatsApp forever. Every Indian services marketplace fights this.

Countermeasures, in order of effectiveness:

1. **Number masking** via cloud telephony (Exotel/Knowlarity) until the booking is confirmed and paid
2. **Make the platform genuinely more useful than the relationship** — tracking link for guests, GST invoice, dispute recourse, guaranteed backup vehicle. These are things a WhatsApp deal cannot provide
3. **Operator-side carrot**: T+2 settlement and free dispatch software, both forfeited on off-platform dealing
4. **Detect it**: flag operators whose quote-to-book ratio drops abnormally, watch for repeat customers who stop returning after one trip with a particular operator
5. **Contractual**: off-platform clause in the operator agreement, enforced with delisting for repeat offenders

Accept that you will lose some. Design the take rate low enough that cheating isn't worth the friction.

Cold start

Classic chicken-and-egg. Solution: **manual, unscalable, city-by-city**. Sign 40 vehicles in one city by walking into their offices. Run the first 200 bookings through a WhatsApp group and a spreadsheet before you write a line of matching code. You will learn more about the quote schema from 50 real negotiations than from six months of design.

Quality variance

A ₹35,000 booking where the wrong vehicle turns up is a catastrophe, not a bad rating. Mitigations: vehicle photos verified against RC, year-of-manufacture displayed, mandatory pre-departure photo by driver, **backup vehicle guarantee** (contract 3–5 float vehicles per city yourself — this is worth the cost), and a compensation policy published upfront.

Seasonality and cash flow

Nov–Feb and Apr–Jun are wedding season; monsoon months are dead. You need working capital to survive Jul–Sep. Model this in the financial plan or you will have a payroll crisis in your first August.

Safety

Group transport means women travelling at night, children on school trips, and elderly pilgrims on mountain roads. One serious incident, handled badly, ends the company. Non-negotiables from day one: SOS with a real 24×7 human on the other end, driver background verification, driving-hours limits enforced in software, night-driving speed alerts, incident escalation runbook, and an insurance policy you've actually read.

11. Roadmap

Phase 0 — Validate without code (Weeks 1–6)

- Incorporate Pvt Ltd; open bank account; engage CA + transport counsel
- Pick **one city** — Jaipur, Delhi NCR, or Bengaluru. Jaipur is the sharpest wedge: enormous wedding + tourism charter demand, dense operator supply, manageable competition
- Sign 25–40 vehicles across 8–12 operators, on paper, with a signed rate card each
- Run **50 real bookings** through WhatsApp + Google Sheets + Razorpay payment links. Charge commission
- Land 2–3 corporate accounts for offsites
- **Output:** a validated quote schema, real rate data, 10 operator relationships, and a clear list of what actually breaks. Do not skip this phase.

Team: 2 founders + 1 ops person. **Cost:** ₹6–10L.

Phase 1 — MVP (Months 2–6)

Ship: customer app (Android + iOS), operator app, driver app, admin console, one city, Lane A (RFQ) only.

Scope discipline — **in:** phone auth, requirement builder, RFQ fan-out, structured quotes, comparison, advance payment via UPI, booking confirmation, driver assignment, basic live tracking, shareable tracking link, GST invoice, ratings, admin verification queue, manual matching desk, settlement runs.

Out: instant book, dynamic pricing, in-app chat (use masked calling), loyalty, referrals, multi-language beyond Hindi/English, corporate SaaS, seat-selling.

Month-by-month:

- **M2:** architecture, design system, auth, entity model, operator onboarding + KYC + VAHAN integration
- **M3:** RFQ engine, quote schema, matching v1 (rules-based), operator app quoting flow
- **M4:** customer app booking flow, payments, invoicing, admin console
- **M5:** driver app, tracking service, tracking link, notifications (WhatsApp), settlement
- **M6:** hardening, load test, security review, store submission, 30-operator closed beta

Team: 1 EM, 3 backend, 2 Flutter, 1 Android, 1 React, 1 designer, 1 QA, 1 DevOps (part-time), 1 PM, 2 ops. **Cost:** ₹85L – ₹1.3Cr including salaries.

Phase 2 — Liquidity (Months 7–11)

- **Lane B instant book** on the top 15 routes, priced from Phase 1 data
- Fair Price band on comparison screen
- Operator rate cards → auto-quoting (targets response rate, the key marketplace metric)
- Empty-leg / return-leg marketplace
- Ratings-driven ranking; operator tiering (Bronze/Silver/Gold) with settlement-speed benefits
- Referrals and wedding-planner partner program
- 3 more cities in the same corridor (Jaipur → Delhi NCR → Agra → Udaipur; a corridor beats scattered cities because vehicles move between them)
- **AIS-140 VLTD feed ingestion**

- iOS parity, 4 more languages

Metric target: 60% quote-response rate within 30 min; 25% RFQ-to-booking conversion; 40% repeat rate for corporates.

Phase 3 — Contracts & scale (Months 12–20)

- **Corporate portal:** cost centres, approval workflows, monthly consolidated billing, SSO, spend analytics
- **Recurring contracts:** employee transport, school routes, hotel tie-ups
- School-trip product with parent tracking access and enhanced driver verification
- ML matching and ETA correction models
- Operator financing partnerships (vehicle loans, fuel cards, insurance) — high-margin, high-retention
- 15–20 cities
- Aggregator licences in each operating state

Phase 4 — Platform (Months 20+)

- Per-seat sale of empty legs (the differentiated version of RedBus, not the imitation)
 - Open API for travel agents, wedding planners, event companies, hotel chains
 - ONDC mobility integration
 - EV fleet incentives, aligned with state EV targets under MVAG 2025
 - Dynamic surge within regulatory bands
-

12. Team and budget

Steady-state team (end of Phase 2)

Function	Count
Engineering Manager / CTO	1
Backend engineers	4
Flutter engineers	2
Android (native, driver app)	1
Frontend (React)	2
Data engineer / analyst	1
QA	1
DevOps / SRE	1
Product Manager	1
Product Designer	1
Product & Engineering	15
City supply managers	1 per city
Ops / booking desk	4–6
Customer support (24×7 rota)	6
Finance / compliance	2
Growth / marketing	2

Indicative budget (INR)

Line	Phase 0–1 (6 mo)	Phase 2 (5 mo)
Engineering salaries	₹75L – ₹1.1Cr	₹1.2Cr – ₹1.6Cr
Ops & supply salaries	₹12L	₹35L
Infrastructure	₹4L	₹12L
Maps, SMS, WhatsApp, telephony	₹3L	₹15L
Legal, CA, licences, deposits	₹8L – ₹15L	₹15L
Marketing & supply incentives	₹10L	₹60L
Backup fleet contracts	₹5L	₹20L
Total	₹1.2Cr – ₹1.6Cr	₹2.8Cr – ₹3.6Cr

A pre-seed of ₹3–4Cr gets you through Phase 2 with runway to raise on real liquidity metrics.

13. Metrics that matter

Marketplace health (watch weekly)

- Quote response rate — % of RFQs receiving ≥ 3 quotes within 30 minutes. *This is the metric. If it falls below 50%, nothing else matters.*
- Time to first quote (target: <10 min)
- RFQ → booking conversion (target: 25%+)
- Supply utilisation — % of listed vehicle-days booked

Operational trust

- On-time vehicle arrival % (target: >95%)
- Booking-to-execution match rate — right vehicle, right driver (target: >98%)
- Operator-initiated cancellation rate (target: <2%)
- SOS/incident rate per 1,000 trips
- Dispute rate and median resolution time

Business

- GMV, take rate, contribution margin per trip
- CAC by channel; LTV by segment
- Repeat rate at 90 days (corporates should exceed 50%)
- Leakage proxy: repeat-customer rate for customers who have used a given operator once

Anti-metrics — resist optimising these

- App downloads (a wedding is booked once a year; installs are a vanity number)
 - Raw operator count (30 responsive operators beat 300 dormant ones)
-

14. Immediate next steps

1. **This week:** trademark search on "Toli" (Class 39), domain and handle registration, MCA name reservation
 2. **Week 2:** incorporate; engage CA (specifically on the §8.3 GST classification question) and transport counsel (on the §8.4 aggregator-licence scope question)
 3. **Weeks 2–6:** run Phase 0 manually in one city. 50 bookings, no code
 4. **Week 4:** hire the EM/CTO; they should be someone who has shipped a marketplace, not just an app
 5. **Week 6:** freeze the quote schema based on what Phase 0 taught you, then start building
 6. **Month 2:** Google Maps Platform India billing account, Razorpay/Cashfree marketplace account, WhatsApp Business API via a BSP, VAHAN/Sarathi API access via an authorised aggregator
 7. **Month 3:** apply for the state aggregator licence — assume 3–6 months of processing
-

Regulatory details in §8 reflect published guidance as of mid-2026 and are summarised for planning purposes. Verify each with your CA and transport counsel before it becomes a line of code or a line of a contract.